

LINKSOCIAL: Linking User Profiles Across Multiple Social Media Platforms

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Abstract—Social media connects individuals to on-line communities through a variety of platforms, which are partially funded by commercial marketing and product advertisements. A recent study reported that 92% of businesses rated social media marketing as very important. Accurately linking the identity of users across various social media platforms has several applications viz. marketing strategy, friend suggestions, multi platform user behavior, information verification etc. We propose LINKSOCIAL, a large-scale, scalable, and efficient system to link social media profiles. Unlike most previous research that focuses mostly on pair-wise linking (e.g., Facebook profiles paired to Twitter profiles), we focus on linking across multiple social media platforms. LINKSOCIAL has three steps: (1) extract features from user profiles and build a cost function, (2) use Stochastic Gradient Descent to calculate feature weights, and (3) perform pair-wise and multi-platform linking of user profiles. To reduce the cost of computation, LINKSOCIAL uses clustering to perform candidate-pair selection. Our experiments show that LINKSOCIAL predicts with 92% accuracy on pair-wise and 74% on multi-platform linking of three well-known social media platforms. Data used in our approach will be available at <http://vishalshar.github.io/data/>.

Index Terms—Social Media Analysis, User Profile Linkage, Social Media Profile Linkage, Entity Resolution

I. INTRODUCTION

Social media is an amalgam of different platforms covering various aspects of an individual's on-line life, such as personal, social, professional, and ideological aspects. For instance, an individual may share professional content on LinkedIn, social pictures on Instagram, and ideas and opinions on Twitter [16]. A recent study found that more than 42% of the adults use more than two social media platforms in everyday life¹.

An individual creates a *profile* to participate in a social media platform. A profile has a *public* view and a *private* view, e.g., a credit card number would be part of the private view. In this paper we are only concerned with a publically available information of a profile that consists of a *username*, *name*, *bio* and *profile image*. This limited profile is at the intersection of the kinds of information in public profiles across social media platforms. An individual has a separate profile for each social media platform.

Social media platforms generate massive amounts of data. Previous studies have analyzed this data to learn a user's behavior [17], interests [18] and recommendations [19]. But

such studies were limited to using only one aspect of an individual's on-line life by harvesting data from a single social media platform. By linking profiles from several platforms it would be possible to construct a much richer body of knowledge about a person and glean better insights about their behavior, social network, and interests, which in turn can help social media providers improve product recommendations, friend suggestions and other services.

User Profile Linkage (UPL) is the process of linking user profiles across social media platforms. Previous research has shown how to use features in a profile to achieve UPL. For instance, 59% of the users prefer to keep their *username* the same across multiple social media platforms [13], which makes the *username* an important feature in UPL. But exploiting such features is not straightforward as there can be inconsistent, missing, or false information between profiles. UPL is also computationally expensive, making it difficult to obtain high accuracy in the linkage across platform [32].

We propose a scalable, efficient, accurate framework called LINKSOCIAL, for linking user profiles on multiple platforms. The framework is depicted in Figure 1. To the left of the figure, LINKSOCIAL collects profiles from Google+, Twitter, and Instagram. The data is cleaned and features of the data are extracted. Next, the similarity is measured in various ways, depending on the feature. Preliminary matches are then refined, and a final match prediction is made. Our empirical evaluation shows the efficacy of LINKSOCIAL.

This paper makes three major contributions.

- 1) We describe how to engineer relevant features for linking user profiles across multiple social media platforms. We show that highly accurate linkage can be achieved by using relatively few public features.
- 2) We show how to decrease the high computation cost of UPL by using clustering. Our intuition is that if we can reduce the number of linkage attempts, then the cost will decrease, so we focus on pruning low similarity linkages. Given a user's profile from one social media platform, to find similar profiles from other platforms, we cluster candidate profiles using similarity based on bi-grams of the *username* and *name*. Our experiments show that optimization preserve accuracy while reducing computation by 90%. The cost reduction claimed does not include pre-computing cost of Jaccard similarity.
- 3) We empirically evaluate the effectiveness of our framework

¹<http://bit.ly/2FiRy8i>

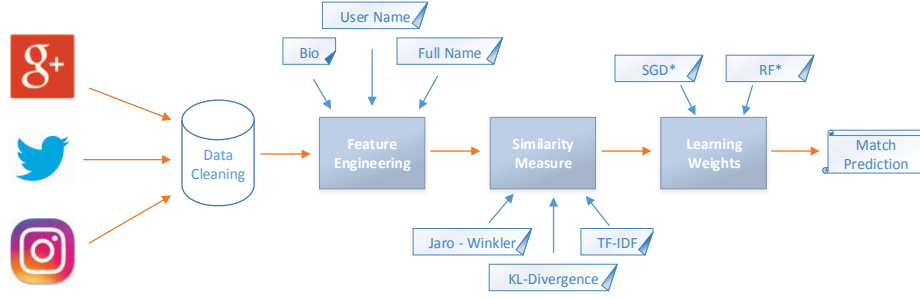


Fig. 1. LINKSOCIAL Framework

by performing extensive experiments and achieving 92% accuracy on pair-wise and 74% accuracy on multi-platform linking.

This paper is organized as follows. Section II motivates the User Profile Linkage problem. In Section III the data acquisition is described. The LINKSOCIAL framework is introduced in Section IV. The section also describes the engineered features, how we measured similarity, and how we optimized for feature weights and reduced the cost of the computation. Section V reports the results of our empirical evaluation of LINKSOCIAL. Related work is presented in Section VI. Section VII presents conclusions and future work.

II. MOTIVATION

In this section, we describe applications of LINKSOCIAL and challenges in UPL.

A. Applications of LINKSOCIAL

Data about an individual's social, professional, personal and ideological aspects can be used in various ways.

Security - Social media is widely used for spreading malicious content [21]. Consider a user spreading such content on a social media platform, their activity can be observed on other platforms using LINKSOCIAL. This can help security agencies identify threats or other malicious activity.

Multi-Platform User Behavior - User behavior and activities have been studied extensively using single social media platforms [17]. Linking behaviors from multiple platforms can create a comprehensive picture of a user's behavior. For example, a user A may be active in social life but disassociate professionally. Understanding multi-platform user behavior may lead to insights into why and how friends network differ across platforms. LINKSOCIAL can help link different behaviors to support multi-platform studies.

Information Verification - A user profile could contain false information. For instance, the mobile social networking app Skout reported that, on average, every two weeks three adults masqueraded as teenagers in its forum [22]. By linking user accounts from multiple social media platforms we can check consistency and improve verification of information by merging and validating information from several sources.

Recommendation - Recommendation of products and services is usually based on data from a single social media platform [19]. Data from multiple platforms can enhance the quality and relevance of recommendations, thus, increases

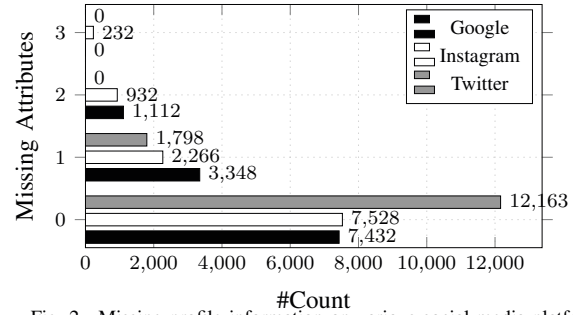


Fig. 2. Missing profile information on various social media platforms.

user engagement. Most friend recommendation algorithms leverage mutual friends. As pointed out by Shu, et al, multi-platform friend recommendations could improve on pair-wise recommendations [20].

B. Challenges in linking users cross social media platforms

Data Collection - Gathering profiles from social media platforms is not trivial since user privacy concerns limit the available information [27]. Even if we could scrape platforms and collect millions of profiles, ground truthing a UPL solution is elusive since there is no independent way to verify that profiles from different platforms belong to the same user.

Incomplete information - The attributes in a public profile differs across platforms. Some platforms may have an e-mail address, provide a location, or contact information, but most do not. Users also incompletely fill out a profile. Some users may expose gender, age, or ethnicity, but not all users due to privacy. The inconsistency of profile attributes between platforms and among users decreases the potential for linkage.

False information - Faking identity on social media is common [23] as is sharing false content [25] and providing false information about an individual [24]. Social media platforms do not provide verification mechanisms for profile data.

Missing information - A profile provides only a small amount of data about a user since privacy concerns limit the amount of public sharing [26]. Profiles with missing data further exacerbates the difficulty of linking as shown in Figure 2. The paucity of available data coupled with the high rate of missing data complicates the task of accurately linking profiles.

Limited Access - Data from major social media platforms can be accessed through a platform-specific API but due to privacy concerns, social media providers reveal only a limited

amount of data. Even after collecting data, we might lack enough common features to link profiles.

III. PROBLEM DEFINITION

This section gives the problem statement, and discusses data collection and pre-processing.

A. Problem Statement.

Let P_i^k be the i^{th} public profile on social media platform k . Let I be an identify function that maps a P_i^k to the identity of the user who created the profile. For linking profiles from n social media platforms, we use the following objective function.

$$\Phi_n(P_i^1, \dots, P_k^n) = \begin{cases} 1 & \text{if } I(P_i^1) = \dots = I(P_k^n) \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Our goal is to build and learn functions $\Phi_2(\cdot)$ and $\Phi_3(\cdot)$ for linking pair-wise and three-platform profiles. We assume that in our dataset, every user has exactly one profile in each social media platform.

B. Dataset Collection

For testing a UPL solution, we need ground truth data. While social media platforms have APIs to access user data, there is no platform to help us link a profile to an individual. However, there are ways that we could build a set of ground truths. For instance, we could crowd-source the ground-truthing (viz. Amazon Mechanical Turks) or use surveys [22], but these methods are prone to getting unreliable data. Instead, we used a novel resource, the website `about.me`, which is a site that requires users to input links to profiles on other social media platforms. When creating a profile on the site, a user will provide links to their other social media profiles.

We used a dataset of 15,298 *usernames* from six social media platforms: Google+, Instagram, Tumblr, Twitter, Youtube, and Flickr [12]. We narrowed the dataset to Instagram, Twitter, and Google+ for our study since they make the *username*, *name*, *bio*, and *profile image* are publicly available.

Next, we built a crawler to collect profiles from the three platforms. Table II displays information about our collected data. We gathered data on 7,729 users from all three platforms, 6,039 users have data available on a pair of platforms and 1,530 have data available only from one platform. Missing profiles could be because of users deactivating their accounts.

C. Dataset Analysis

We analyzed the collected profiles to determine how much information was missing. Figure 2 shows the count of missing profile attributes for each platform. In Google, Twitter, and Instagram there were 28%, 13% and 21% of user profiles with at least one missing attributes, respectively. 9% and 8% of Google and Instagram profiles had at least two missing attributes. There were 87%, 62%, and 69% of profiles on Twitter, Google, and Instagram, respectively, without missing information. Three attributes were missing from 2% of Instagram profiles, making it impossible to match them. The attributes that were presented have some variance. On average,

TABLE I
NUMBER OF PROFILES OBTAINED PER SOCIAL MEDIA PLATFORM.

Social Media	Profile Count
Instagram	10,958
Twitter	13,961
Google+	11,892

TABLE II
NUMBER OF USERS WITH PROFILES ON PAIR-WISE AND MULTI-PLATFORMS

Social Media	Profile Count
Instagram – Google+	614
Twitter – Instagram	2451
Google+ – Twitter	2974
Google+ – Instagram – Twitter	7729

^aThere were 1530 users with only one profile.

bios on Google+ were longer than Instagram and Twitter; 164 characters on Google+ compared to 70 and 96 for Instagram and Twitter. However, the *username* attribute has little variance, it was 11-13 characters on average across all three platforms.

IV. THE LINKSOCIAL FRAMEWORK

This section describes LINKSOCIAL, discusses computation cost, and shows how to reduce the cost using clustering.

A. Feature Engineering

LINKSOCIAL uses the basic features in a profile as well as the following engineered features.

Username and Name bi-grams - The *username* is an important feature for profile linkage since people tend to use the same *username* across social media platforms. When a new name is used, parts of the name is often re-used. For example, John_snow, johnsno, and snow_john could belong to the same person. String similarity metrics such as Jaro-Winkler, longest common subsequence, or Levenshtein distance, tend to perform poorly on *name* transpositions [34]. To better align names, we engineer the bi-grams of *usernames* as a feature. We also merge the bi-grams of *usernames* with the bi-grams of names as a feature since people also like to transpose their surname and first name in for a *username*. We engineer the following feature sets.

- bi-gram of *username*. (u_u)
- bi-gram of *name*. (u_n)
- merging above two features. (u_b)

These three feature sets capture a range of different ways to create a *username*.

Character Distribution - User's like to create *usernames* using substrings of their *name* or other personal information (a pet's name or a significant date). To handle scenarios where bi-grams could not capture the similarity, we use the character distribution of *usernames* and *names* as features. To measure distribution similarity we use Kullback-Leibler divergence as defined below,

$$KL_{divergence}(P||Q) = \sum_{i=1}^n P(i) \cdot \log \frac{P(i)}{Q(i)} \quad (2)$$

where P and Q are given probability distributions. We engineered the following features sets using character distribution similarity.

- *username*. (u_{u_sim})
- *name*. (u_{n_sim})
- *username + name*. (u_{b_sim})

We perform experiments on real data and have not considered scenarios where *username* and *name* have no relationship.

Profile Picture Similarity - Users often use the same profile picture in multiple platforms. To capture similarity between *profile images*, we use OpenFace [14], an open source image similarity framework. Openface crops images to extract face and uses deep learning to represent the face on a 128-dimensional unit hypersphere. We use ℓ_2 norm to calculate distance between vectors of two *profile images*.

B. Similarity Measures

LINKSOCIAL uses two similarity measures.

Jaro-Winkler Distance - Jaro-Winkler is metric for string matching and is commonly used when matching *names* in UPL [11]. Studies show that the metric performs well for matching string data [15]. Jaro accounts for insertions, deletions, and transpositions while Winkler improves Jaro based on the idea that fewer errors occur early in a *name*.

TF-IDF and Cosine Similarity - LINKSOCIAL uses a different similarity measure for matching *profile bios* since *bios* are longer than *names*. TF-IDF and cosine similarity are widely used for measuring the similarity of documents.

C. Matching Profiles

LINKSOCIAL matches profiles using the basic and engineered features of a profile. We transform the matching problem into a multivariate binary classification problem and optimize it using Stochastic Gradient Descent (SGD). It is defined as follows:

$$h(x) = w_0 + w_1 \cdot x_1 + w_2 \cdot x_2 + w_3 \cdot x_3 + \dots + w_m \cdot x_m \quad (3)$$

In the equation, $x_1, x_2, \dots, x_m$ represents the similarity score of features between two profiles and $w_1, w_2, \dots, w_m$ represents their respective weights or coefficient. We use Mean Squared Error (*MSE*), as our loss function. Considering the predicted values as $h_w(x)_{(i)}$ where $i \in 1, 2, \dots, n$ and $y_{(i)}$ as a given value (either a match (1) or no match (0)), we define *MSE* or cost function $f_{cost}(W)$ as follows:

$$MSE = f_{cost}(W) = \frac{1}{M} \sum_{i=1}^m (y_{(i)} - h_w(x)_{(i)})^2 \quad (4)$$

LINKSOCIAL uses SGD to optimize the cost function. Partial derivatives of Equation 4 w.r.t to w_1 & w_2 are defined as follows:

$$\begin{aligned} \nabla_{w_1} f'_{cost}(w_1) &= \frac{1}{M} \sum_{i=1}^m -2x_1 (y_{(i)} - h_w(x)_{(i)}) \\ \nabla_{w_2} f'_{cost}(w_2) &= \frac{1}{M} \sum_{i=1}^m -2x_2 (y_{(i)} - h_w(x)_{(i)}) \end{aligned}$$

and similarly for other weights.

Derivatives are followed by updating of the weights. For example, w_1 & w_2 as shown below:

$$\begin{aligned} w_1 &= w_1 - \eta \nabla_{w_1} f'_{cost}(w_1) \\ w_2 &= w_2 - \eta \nabla_{w_2} f'_{cost}(w_2) \end{aligned}$$

In the above equation, η is the learning rate, a value that typically ranges between 0.0001 - 0.005. During experiments, we also add elastic net regularization for training.

The derivatives and weights are recursively calculated until the equation converges and yields an optimized weight for each attribute based on a training set. To find a match of a given profile we use Equation 5 where we find the profile with maximum score on the given attributes and weights. In Equation 5, W^T is a weight vector calculated using the optimization of Equation 4 and X_u is a vector of attributes. P_i^k is a profile from social media platform k , while U^j is a set of all profiles from platform j .

$$Match(P_i^k, U^j) = \max(W^T \cdot X_u), \quad \forall u \in U^j \quad (5)$$

Given profile P_i^k , *Match()* outputs the most similar profile from platform j .

D. Computation Reduction Using Clustering

UPL can be computationally expensive. Given our dataset with 7,729 user profiles, if we are linking pairs of profiles from only two of the platforms, then the number of comparisons will be $7,729 * 7,728 = 59,729,712$. Assuming we can perform 1,000 comparisons per second (which is a very high ballpark), it will require 17 hours to perform UPL. Matching of millions of users across multiple platforms will be infeasible (without the dedication of massive computing power).

To tackle this problem, we introduce *candidate profile clustering*. We can reduce the number of comparisons by pruning low potential comparisons, that is, by avoiding the work of matching profiles that are dissimilar. In our dataset, 45% of the Instagram and Twitter profiles have the same *username* and *name*. By clustering on the bi-gram features for *username* and *name* we can prune comparisons from profiles in different clusters. We have observed in previous studies [35], Jaccard similarity relatively performs well for finding similarity between bi-grams/n-grams and is also computationally not expensive. We rank profiles based on Jaccard similarity and we choose the top 10% of the candidate profiles with the highest score in the cluster. Algorithm 1 gives our approach for building clusters. Our clustering approach is conceptually similar to kNN clustering where distance is defined by Jaccard similarity j_sim and k is 1.

Clustering of profiles before linking can help reduce computation cost significantly but *how big should our clusters be?* If the cluster is big, the computation cost will increase; if cluster is small, we may fail to capture maximum profile matches. To understand the effect of clustering, we plotted cluster size (as a percentage of the total number of profiles) versus match accuracy on our training data as shown in Figure 3. In the plot, the x -axis is the percentage of profiles in a cluster (on

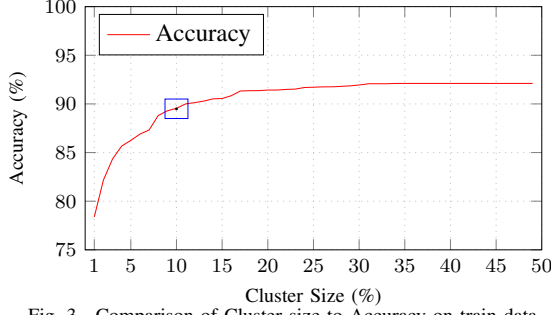


Fig. 3. Comparison of Cluster size to Accuracy on train data.

average). The y -axis shows the maximum accuracy we can achieve since the matching profile has to be in the cluster. We observed that clusters roughly 10% of the size of the data reduces 90% of computation cost while preserving $\sim 90\%$ of the potential matches.

Algorithm 1 Computing candidate profiles for a given user u

Input: $u, \mathcal{U}$ $\triangleright \mathcal{U}$ is set of all users from a different social media than u .

- 1: **procedure** FIND_CANDIDATE_PROFILE($u, \mathcal{U}, n$) $\triangleright n$ is number of profile to find in cluster.
- 2: $u_{bg} \leftarrow u_u + u_n$
- 3: **for each** $p \in \mathcal{U}$ **do** $\triangleright p$ is single user profile
- 4: $p_{bg} \leftarrow p_u + p_n$
- 5: $j_sim \leftarrow Jaccard_similarity(u_{bg}, p_{bg})$
- 6: $Score[p] \leftarrow j_sim$ $\triangleright$ Insert in dictionary
- 7: **end for**
- 8: $Candidate_Profiles = get_top_n(n, Score)$
- 9: **return** $Candidate_Profiles$ $\triangleright$ Returns n candidate profile similar to u
- 10: **end procedure**
- 11: **procedure** JACCARD_SIMILARITY(u_{bg}, p_{bg})
- 12: $Intersection = |u_{bg} \cap p_{bg}|$ $\triangleright$ Number of common elements.
- 13: $Union = |u_{bg} \cup p_{bg}|$ $\triangleright$ Number of unique elements.
- 14: $Jaccard_Similarity = \frac{Intersection}{Union}$
- 15: **return** $Jaccard_Similarity$
- 16: **end procedure**
- 17: **procedure** GET_TOP_N($n, Score$)
- 18: $Score \leftarrow Sort(Score)$ $\triangleright$ Sort Score w.r.t value
- 19: $top_key = get_key(Score, n)$ $\triangleright$ returns top n key from sorted Score
- 20: **for each** $key \in top_key$ **do**
- 21: $top_n \leftarrow Score[key]$
- 22: **end for**
- 23: **return** top_n $\triangleright$ return top n candidate profiles
- 24: **end procedure**

V. EXPERIMENTS

This section reports an experimental evaluation of LINKSOCIAL. We establish a baseline for UPL and verify that LINKSOCIAL can learn Equation 5 and achieve high matching accuracy. We use several variations of calculating weights and compare them with the baseline to determine the impact of the weight calculations. Specifically, we use Random Forest (RF) and Stochastic Gradient Descent (SGD) for calculating weights. We also performed feature analysis to understand which features are important.

TABLE III
COMPARISON WITH PREVIOUS RESEARCH

Linkage	Authors et. al	Features	Dataset Public	Scalable	Reduce Cost	Scalable Across Platform
Pair-Wise	P. Jain[28]	Private	×	×	×	×
	A. Mal[8]	Public	×	✓	✓	×
	R. Zafa[22]	Public	×	×	×	×
	Y. Li[31]	Private	×	×	×	×
	LINKSOCIAL	Public	✓	✓	✓	✓
ACROSS	X. Mu[32]	Private	×	×	×	✓
	S. Liu[6]	Private	×	×	×	✓
	LINKSOCIAL	Public	✓	✓	✓	✓

A. Experimental Setup

We used the data described in Section III-B. We chose a 60-40 split in the data for training and testing.

Baseline - We built a baseline using Jaro-Winkler and TF-IDF as discussed in Section IV-B. We use Jaro-Winkler to analyze *username* and *name* similarity, and TF-IDF and cosine similarity for profile bios. To find the match of a user profile, we select a profile with the highest score from Equation 6 where the value of the W vector is 1, considering each feature is equally important and X represents the feature vector.

$$f(W) = W^T \cdot X_u \quad (6)$$

Calculating Weights - To calculate weights for pair-wise linking, we generated all the features discussed in Section IV-A for each pair, which gives us data for correct matches. To generate data for mismatches, we randomly chose pairs equal to the number of correct matches and collected their feature scores.

We used normalized variable importance score from RF and also SGD optimization algorithm as explained in Section IV-C for weights calculation. RF was trained using 10-Fold Cross Validation, it was tuned using grid search, and the square root of the number of features was selected as the number of variables to be randomly sampled as candidates at each split. SGD was also performed on the same dataset to calculate weights with a learning rate η of 0.001, with 1,000 iterations.

Computing Candidate Profile - To compute candidate pairs, we follow the approach described in Algorithm 1. In training, we pre-compute each profile's potential matches. To generate a feature vector we score based on clusters. To make sure we have both positive and negative label samples of equal size, we randomly sample negative label data of the same size as the positive label set and generated samples are used to learn the weight vector.

Multi-Platform Profile Linking - We link users across three platforms similar to how we perform pair-wise linking. Due to the very high computation cost, we were unable to run experiments for linking without first clustering candidate profiles. We performed experiments by adding and removing engineered features. To find a similar profile, we choose a profile in one social media platform and tried to find the matching profiles in the other two platforms. We performed

TABLE IV
LINKSOCIAL PERFORMANCE ON PAIR-WISE UPL

	Social Media Pairs (Accuracy %)		
Experiments	G+≡I	T≡I	G+≡T
<i>baseline</i>	55.36%	77.86%	56.86%
<i>Prediction without engineered features and clustering.</i>			
with RF	77.53%	82.08%	77.14%
with SGD	76.61%	82.21%	66.24%
<i>Prediction with clustering, no engineered features.</i>			
with CP & RF	82.62%	83.33%	81.40%
with CP & SGD	82.47%	83.32%	81.19%
<i>Prediction with engineered features, no clustering.</i>			
with RF	86.54%	91.17%	84.56%
with SGD	86.63%	91.68%	84.58%
<i>Prediction with engineered features and clustering.</i>			
with CP & RF	84.85%	87.92%	83.20%
with CP & SGD	84.91%	88.29%	83.23%

this experiment for each platform. During training, we choose a platform and compute its respective candidate profiles from other platforms by building feature vector between candidate profiles and the given profile. We then perform SGD and RF to calculate weights for each feature and we used the calculated weights to find similar profiles.

Previous research comparison - Table III compares our approach to previous approaches in terms of feature selection, data availability and scalability. Previous approaches have used different types of features in their algorithms. A *public feature* is publicly available information (e.g., *bio*, *profile image*, *name*, *user name*). *Private* data such as *location* data is proprietary limiting the feasibility of approaches that rely on such information since they need cooperation from businesses that compete against each other. Making a publicly available benchmark dataset is an important aspect for future work and comparison of approaches. Previous researchers have not shared their data (because of privacy concerns and data sources), where as in our approach we have collected all data as public information and have made it available for future research/comparison. As discussed earlier, UPL is a computationally expensive process and scalability of an approach depends on using methods to reduce computation cost to make the solution practical in a real-life scenario. Also, designing an algorithm to scale for new/several platforms is very important aspect. Only a few approaches in past have used computation cost reduction and are designed to scale for new social media platforms.

Previous work accuracy Table VI reports accuracies in a sampling of previous research in both pair-wise and multi-platform linkage. The accuracy is reported “as is” from the papers, experimental setups and measurements differ across papers, for instance previous work with better accuracy for pair linkage used user generated content data (gaining access to such data is difficult) but in our approach we use only publicly available profile data We observe that LINKSOCIAL is among the leaders in pair-wise UPL and the best at multi-platform UPL.

TABLE V
LINKSOCIAL PERFORMANCE ON MULTI-PLATFORM UPL

	Cross Platform		
Experiments	T→(G+, I)	G+→(T, I)	I→(G+, T)
CP & RF	71.56%	72.50%	73.70%
CP & SGD	72.95%	72.86%	74.18%

*RF—Random Forest, SGD—Stochastic Gradient Descent, CP—Candidate Profiles using Clustering, T—Twitter, G+—Google+, I—Instagram

TABLE VI
REPORTED ACCURACY FROM FEW PREVIOUS WORK.

Linkage	Authors	Accuracy
Pair-Wise Social Platform Linkage	P. Jain et al. [28]	39.0%
	A. Malhotra et. al. [8]	64.0%
	R. Zafarani et. al. [22]	93.8%
	Y. Li et. al. [31]	89.5%
	Our Approach	91.68%
Across Multiple Platform	X. Mu et. al. [32]	44.00%
	S. Liu et. al. [6] (reported by [32])	42.00%
	Our Approach	74.18%

B. Evaluation Metrics

In previous studies, accuracy has been used as a reliable evaluation metric for UPL [28]. Given two profiles from different platforms, the accuracy of such matching can be measured as follows. First, assume the following are known.

- *Total number of correct prediction (P)*: Number of correct positive prediction by LINKSOCIAL.
- *Total number of positive sample (N)*: Number of positive linked profiles in the dataset.

Then the accuracy can be computed as follows.

$$Accuracy(\%) = \frac{|P|}{|N|} \cdot 100 \quad (7)$$

C. Results

We performed several experiments on pair-wise linking and the results are shown in Table IV and Table V. Specifically, we measured the accuracy of LINKSOCIAL on all possible pairs in our dataset namely, Google+ - Twitter, Google+ - Instagram, and Twitter - Instagram. We also performed experiments with features and weights calculated using RF and SGD. As shown in Table IV, we started with building a baseline for each pair. We achieved 55%, 78%, 57% for Google+ - Instagram, Twitter - Instagram and Google+ - Twitter respectively. We then performed experiments without using engineered features and clustering. We observed that RF produced more accurate matches than SGD. Next, we added candidate pairs but still no engineered features. In this case, both RF and SGD performed equally well. We then added engineered features and perform experiments without clustering. We observed that SGD outperformed RF. Finally, we used both engineered features and clustering. SGD again performed better than RF. Overall, weights calculated using SGD proved to be more accurate than RF though in some cases the difference was marginal. In the final stage, we observed reduction of accuracy by adding clustering. This is due to the inconsistent *username* and *name* used by a user, since clustering uses *username* and *name* (in our dataset, out of all

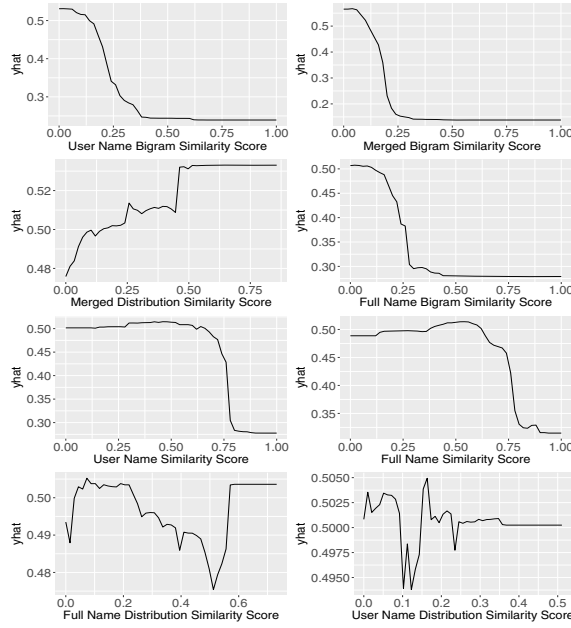


Fig. 4. Partial Dependence Plots for individual features.

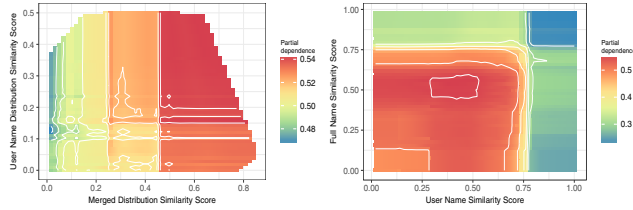


Fig. 5. Partial Dependence Plots for selected pairs of features.

pairs, maximum of 45% users have the same *username* and 22.2% of users with the same *name*), but slight decrease in accuracy could be traded for speed.

We performed several experiments on multi-platform linking and the results are shown in Table V. We observed that, feature weights computation using SGD again outperformed RF. We achieved an accuracy of 73%, 73%, 74% for Twitter→(Google+, Instagram), Google+→(Twitter, Instagram), and Instagram→(Twitter, Google+) respectively. Overall, in our experiments on multi-platform linkage, SGD proved to be more accurate.

Model Interpretation - To understand our model, we performed feature analysis using Partial Dependence Plots (PDP) [29]. Given predictor variables and response variables, PDP can interpret the relationship between them. We start by studying the effect of all variables and later choose variable pairs for further study.

In Figure 4, the x -axis represents feature similarity score and the y -axis represents the effect on the class probability. In Figure 4 we observed, that the score from the merged distribution similarity is positively correlated to the model, *username* and *name* similarity contributes to the model until the value of 0.75 then it drops. We plot selected variable pairs, to study their effect on the model with the results show in Fig-

ure 5. In Figure 5, the x -axis and y -axis represent the score of respective feature similarity and Partial Dependence represents the marginal effect of features on the class probability. We observed that higher values of merged and *username* distribution similarity score together are highly correlated to the model. Similarly, *username* and *name* similarity score values until 0.75 are highly correlated but the highest values are relatively low. This implies that there are several instances in our data where *username* and *name* similarity scores together are very high (close to 1), but selected profiles do not belong to the same individual. We also observed, with their relatively lower values of similarity, that there are instances where both profiles belong to the same person. Finally, we concluded that the similarity of *username* and *name* are insufficient or unreliable features for linking profiles.

VI. RELATED WORK

UPL is a subproblem of a larger problem that has been studied under different names such as record linkage, entity resolution, profile linkage, data linkage, and duplicate detection. In the field of databases, *entity resolution* links an entity in one table/database to another entity from another table/database, *e.g.*, when linking data from the healthcare to the insurance data [2]. Entity resolution has been referred to as *coreference resolution* [1] in NLP and *named disambiguation* [3] in IR. Approaches to solving the problem fall into three categories: numerical, rule and workflow-based [20]. Numerical approaches use weighted sums of calculated features to find similarity. Rule-based approaches match using a threshold on rule for each feature. Workflow-based approaches use iterative feature comparisons to match.

There have been several approaches that utilize user behavior to solve the pair-wise matching problem, such as, a model based approach [6], a probabilistic approach [5], a clustering approach [7], behavioral approach [28], user generated content [31], and both supervised [9] and unsupervised learning approaches [10]. The problem of user linkage on social media was formalized by Zefarani et al. [13] where they used usernames to identify the corresponding users in different social community. In our framework, we used a supervised approach, and mitigate cost by reducing the number of comparisons.

Most previous work in UPL focuses on pair-wise matching due to challenges in computational cost and data collection. In pair-wise matching, Qiang Ma et al. [4] approached the problem by deriving tokens from features in a profile and used regression for prediction; R. Zefarani et al. [22] used username as a feature and engineered several other features by applying a supervised approach to the problem. Unlike these approaches, LINKSOCIAL can perform multi-platform matching as well as pair-wise matching. Multi-platform UPL has received less attention. Xin et. al [32] approached the multi-platform UPL using latent user space modeling, Silvestri et.al [33] uses attributes, platform services, and matching strategies to link users on Github, Twitter, and StackOverflow; Gianluca et. al [30] leverages network topology for matching profiles

across n social media. Liu et. al. [6] uses heterogeneous user behavior (user attributes, content, behavior, network topology) for multi-platform UPL but gaining access to such data is not a trivial task.

VII. CONCLUSION

In this paper, we investigate the problem of User Profile Linkage (UPL) across social media platforms. Multi-platform linkage can provide a richer, more complete foundation for understanding a user's on-line life and can help improve several research studies currently performed only on single social media platform. UPL has many potential applications but is challenging due to the limited, incomplete, and potentially false data on which to link. We proposed a large scale, efficient and scalable solution to UPL which we call LINKSOCIAL. LINKSOCIAL links profiles based on a few core attributes in a public profile: *username*, *name*, *bio* and *profile image*. Our framework consists of (1) feature extraction, (2) computing feature weights, and (3) linking pair-wise and multi-platform user profiles. We performed extensive experiments on LINKSOCIAL using data collected from three popular social media platforms: Google+, Instagram and Twitter. We observed that *username* and *name* alone are an insufficient set of features for achieving highly accurate UPL. UPL is computationally costly, but we showed how to use clustering to reduce the cost without sacrificing accuracy. Candidate profile clustering is based on pruning dissimilar profile comparisons. It reduced 90% of the comparisons which significantly helped in scaling our framework. We evaluate our framework on both pair-wise and multi-platform profile linkage with accuracy 91.68% on pair-wise and 74.18% on multi-platform linkage.

Data about a user from multiple social media platforms has many applications. In future, we plan to (1) extend our work to study a user's behavior across platforms, which to our knowledge has not yet been studied, (2) add more features to LINKSOCIAL using heterogeneous data, *e.g.*, user content similarity (text, videos, images), network similarity, and patterns across social media platforms, and (3) evaluate LINKSOCIAL on more (up to six) social media platforms.

REFERENCES

- [1] J. Cai, M. Strube, "End-to-End Coreference Resolution via Hypergraph Partitioning," COLING, 2010.
- [2] S. E. Whang, D. Marmaros, H. Garcia-Molina, "Pay-As-You-Go Entity Resolution," IEEE Transactions on Knowledge and Data Engineering, 2013.
- [3] Y. Qian, Y. Hu, J. Cui, Q. Zheng, Z. Nie, "Combining machine learning and human judgment in author disambiguation," CIKM, 2011.
- [4] Q. Ma, H. H. Song, S. Muthukrishnan, A. Nucci, "Joining user profiles across online social networks: From the perspective of an adversary," IEEE/ACM ASONAM, 2016.
- [5] H. Zhang, M. Kan, Y. Liu, S. Ma, "Online Social Network Profile Linkage," AIRS, 2014.
- [6] S. Liu, S. Wang, F. Zhu, J. Zhang, R. Krishnan, "HYDRA: large-scale social identity linkage via heterogeneous behavior modeling," SIGMOD Conference, 2014.
- [7] W. W. Cohen, J. Richman, "Learning to match and cluster large high-dimensional data sets for data integration," KDD, 2002.
- [8] A. Malhotra, L. C. Totti, W. Meira Jr., P. Kumaraguru, V. A. F. Almeida, "Studying User Footprints in Different Online Social Networks," 2012 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining, 2012.
- [9] A. Nunes, P. Calado, B. Martins, "Resolving user identities over social networks through supervised learning and rich similarity features," SAC, 2012.
- [10] J. Liu, F. Zhang, X. Song, Y. Song, C. Lin, H. Hon, "What's in a name?: an unsupervised approach to link users across communities," WSDM, 2013.
- [11] W. E. Winkler, "Overview of record linkage and current research directions," BUREAU OF THE CENSUS, 2006.
- [12] B. Lim, D. Lu, T. Chen, M. Kan, "# mytweet via instagram: Exploring user behaviour across multiple social networks," Advances in Social Networks Analysis and Mining (ASONAM), 2015 IEEE/ACM International Conference on, 2015.
- [13] R. Zafarani, H. Liu, "Connecting Corresponding Identities across Communities," ICWSM, 2009.
- [14] T. Baltrušaitis, P. Robinson, L. Morency, "Openface: an open source facial behavior analysis toolkit," Applications of Computer Vision (WACV), 2016 IEEE Winter Conference on, 2016.
- [15] W. Cohen, P. Ravikumar, S. Fienberg, "A comparison of string metrics for matching names and records," Kdd workshop on data cleaning and object consolidation, 2003.
- [16] L. Manikonda, V. V. Meduri, S. Kambhampati, "Tweeting the Mind and Instagramming the Heart: Exploring Differentiated Content Sharing on Social Media," ICWSM, 2016.
- [17] F. Benevenuto, T. Rodrigues, M. Cha, V. A. F. Almeida, "Characterizing user behavior in online social networks," Internet Measurement Conference, 2009.
- [18] P. Bhattacharya, M. BilalZafar, N. Ganguly, S. Ghosh, K. P. Gummadi, "Inferring user interests in the Twitter social network," RecSys, 2014.
- [19] M. Jamali, M. Ester, "A matrix factorization technique with trust propagation for recommendation in social networks," RecSys, 2010.
- [20] K. Shu, S. Wang, J. Tang, R. Zafarani, H. Liu, "User Identity Linkage across Online Social Networks: A Review," SIGKDD, 2016.
- [21] J. Klausen, "Tweeting the Jihad: Social media networks of Western foreign fighters in Syria and Iraq," Studies in Conflict & Terrorism, 2015.
- [22] R. Zafarani, H. Liu, "Connecting users across social media sites: a behavioral-modeling approach," KDD, 2013.
- [23] S. C. Herring, S. Kapidzic, "Teens, Gender, and Self-Presentation in Social Media," KDD, 2014.
- [24] G. S. O'Keeffe, K. Clarke-Pearson, "The impact of social media on children, adolescents, and families," Pediatrics, 2011.
- [25] D. Miller, E. Costa, N. Haynes, T. McDonald, R. Nicolescu, J. Sinanan, J. Spyer, S. Venkatraman, X. Wang, "How the world changed social media," UCL press, 2016.
- [26] R. Gross, A. Aless, "Information revelation and privacy in online social networks," WPES, 2005.
- [27] J. M. Kleinberg, "Challenges in mining social network data: processes, privacy, and paradoxes," KDD, 2007.
- [28] P. Jain, P. Kumaraguru, A. Joshi, "@I Seek 'Fb.Me': Identifying Users Across Multiple Online Social Networks," WWW, 2013.
- [29] A. Goldstein, A. Kapelner, J. Bleich, E. Pitkin, "Peeking inside the black box: Visualizing statistical learning with plots of individual conditional expectation," journal of Computational and Graphical Statistics, 2015.
- [30] G. Quercini, N. Bennacer, M. Ghufan, C. NanaJipmo, "LIAISON: reconciliation of Individuals Profiles Across SOcial Networks," Advances in Knowledge Discovery and Management: Volume 6, 2017.
- [31] Y. Li, Z. Zhang, Y. Peng, H. Yin, Q. Xu, "Matching user accounts based on user generated content across social networks," Future Generation Computer Systems, 2018.
- [32] X. Mu, F. Zhu, E. Lim, J. Xiao, J. Wang, Z. Zhou, "User Identity Linkage by Latent User Space Modelling," Proceedings of the 22Nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016.
- [33] G. Silvestri, J. Yang, A. Bozzon, A. Tagarelli, "Linking Accounts across Social Networks: the Case of StackOverflow, Github and Twitter," KDWeb, 2015.
- [34] P. Christen "A Comparison of Personal Name Matching: Techniques and Practical Issues," Workshops Proceedings of the 6th IEEE ICDM, 2006.
- [35] M. Krieger and D. Ahn "TweetMotif: exploratory search and topic summarization for Twitter," In Proc. of AAAI Conference on Weblogs and Social, 2010